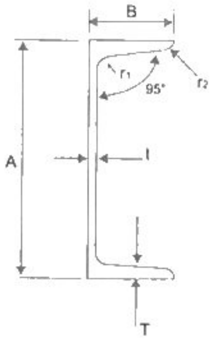
Mild Steel Channels, Flats, Angles and Square Bars

Type	Specification	Chemical Composition (Ladle Analysis)			Tensile Properties						Bend Test		
					Yield Stress (MN/m^2)		Tensile Strength (MN/m^2)	Eongation			Bending Angle	Inside radius	Test Piece
								t<5mm	5<t<16mm	16<t<50mm			
		C%	P%	S%	t<16mm	16<t<40mm		Test Pc No 5	Test Pc No 1A	Test Pc No 1A			
Flat Bar	JIS G3101 1976, Class 2 SS 400	–	0.050 max.	0.050 max.	245 min.	235 min.	402 to 510	21% min.	17% min.		180°	1.5t*	No.1
Angle Bar													
Square Bar													

Note: t* = Thickness of flat, angle and square bar

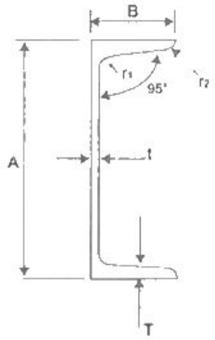
Channels



METRIC SERIES

Size	Dept (A)		Flange Width (B)		Web (t) Thickness		Flange (T) Thickness		Corner Radius				Sectional Area	
A X B X t	mm	in	mm	in	mm	in	mm	cm	mm	in	mm	in	cm	in
380 x 100 x 10.5	380	14.961	100	3.937	10.5	.431	16	.630	18	.709	9	.354	69.39	10.76
380 x 100 x 13	380	14.961	100	3.937	13	.512	6.5	.650	18	.709	9	.354	78.96	12.54
300 x 90 x 9	300	11.811	90	3.543	9	.354	12	.512	14	.551	7	.276	48.57	7.531
300 x 90 x 10	300	11.811	90	3.543	10	.394	15.5	.611	19	.748	9.5	.374	55.74	8.640
280 x 100 x 9	280	11.024	100	3.937	9	.354	13	.512	14	.551	7	.276	49.37	7.652
280 x 100 x 11.5	280	11.024	100	3.937	11.5	.453	16	.630	18	.709	9	.354	61.37	9.512
250 x 80 x 8	250	9.843	80	3.150	8	.315	12.5	.492	14	.551	7	.276	38.51	5.969
250 x 90 x 9	250	9.843	90	3.543	9	.354	13	.512	14	.551	7	.276	44.07	6.831
250 x 90 x 11	250	9.843	90	3.543	11	.433	14.5	.571	17	.669	8.5	.335	51.17	7.931
230 x 80 x 8	230	9.055	80	3.150	8	.315	12.	.472	13	.512	6.5	.256	36.12	5.599
230 x 90 x 8.5	230	9.055	90	3.543	8.5	.335	13.5	.531	15	.591	7.5	.295	42.14	6.532
200 x 80 x 7.5	200	7.874	80	3.150	7.5	.295	11	.433	12	.472	6	.236	31.33	4.856
200 x 90 x 8	200	7.874	90	3.543	8	.315	13.5	.531	14	.551	7	.276	38.65	5.991
180 x 75 x 7	180	7.087	75	2.953	7	.276	10.5	.413	11	.433	5.5	.217	27.20	4.216
180 x 90 x 7.5	180	7.087	90	3.543	7.5	.295	12.5	.492	13	.512	6.5	.256	34.57	5.358
150 x 75 x 6.5	150	5.906	75	2.953	6.4	.256	10	.394	10	.394	5	.197	23.71	3.675
150 x 75 x 9	150	5.906	75	2.953	9	.354	12.5	.492	15	.591	7.5	.295	30.59	4.741
125 x 65 x 6	125	4.921	65	2.559	6	.236	8	.315	8	.315	4	.157	17.11	2.652
100 x 50 x 5	100	3.937	50	1.969	5	.197	7.5	.295	8	.315	4	.157	11.92	1.848
75 x 40 x 5	75	2.953	40	1.575	5	.197	7	.276	8	.315	4	.157	8.818	1.367

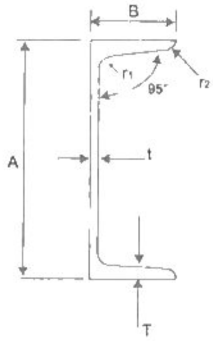
Channels



METRIC SERIES

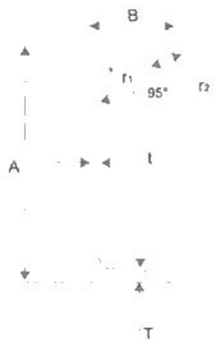
Weight			Centre of Gravity		Moment of inertial				Radius of Gyration				Modulus of Section				Size
kg/m	kg/ft	lb/ft	cm	in	cm	in	cm	in	cm	in	cm	in	cm	in	cm	in	A x B x t
54.5	16.6	36.60	2.41	.949	14500	348.4	557	13.38	14.5	5.709	2.83	1.114	782	46.50	73.3	4.473	380 x 100 x 10.5
62.0	18.9	41.66	2.29	.902	15600	374.8	584	14.03	14.1	5.551	2.72	1.071	822	50.16	78.8	4.625	380 x 100 x 13
38.1	11.6	25.57	2.23	.878	6440	154.7	325	7.808	11.5	4.528	2.59	1.020	429	26.18	48.0	2.929	300 x 90 x 9
43.8	13.3	29.32	2.33	.917	7400	177.8	373	8.961	11.5	4.528	2.59	1.020	494	30.15	56.0	3.417	300 x 90 x 10
38.8	11.8	26.07	2.64	1.040	5930	142 5	428	10.28	11.0	4.331	2.95	1.161	423	25.81	58.2	3.552	280 x 100 x 9
48.2	14.7	32.39	2.68	1.055	7150	171.8	515	12.37	10 8	4.252	2.90	1 142	510	31.12	70.4	4.296	280 x 100 x 11.5
30.2	9.22	20.29	2.11	.831	3630	87.21	210	5.045	9.71	3.823	2.34	921	291	17.76	35.7	2.179	250 x 80 x 8
34.6	10.5	23.15	2.42	953	4180	100 4	306	7 352	9 74	3.835	2.64	1.039	335	20 44	46.5	2.838	250 x 90 x 9
40.2	12.2	26.90	2 39	.941	4690	112.7	342	8.217	9.57	3.768	2.58	1.016	375	22.88	51.7	3.155	250 x 90 x 11
28.4	8.64	19.08	2 15	.846	2900	69.67	200	4.805	8.96	3.528	2.35	.925	252	15.38	34.2	2.087	230 x 80 x 8
33.1	10.1	22.24	2 58	1.016	3490	83.85	303	7.280	9.10	3 583	2.68	1.055	304	18.55	47.3	2 886	230 x 90 x 8.5
24.6	7.50	16.53	2.24	.884	1950	46.85	177	4.525	7 89	3 106	2.38	.937	195	11.90	30.8	1.880	200 x 80 x 7.5
30.3	9.25	20.40	2.77	1.091	2490	59.82	286	6.872	8.03	3 161	2.72	1.071	249	15 19	45.9	2.801	200 x 90 x 8
21.4	6.51	14.35	2 15	846	1380	33 15	137	3.291	7.13	2 807	2 24	882	154	9 398	25 5	1 556	180 x 75 x 7
27.1	8.27	18.21	2.85	1.122	1840	44.21	258	6.198	7.29	2.870	2.73	1.075	204	12.45	42.0	2.563	180 x 90 x 7.5
18.6	5.67	12.50	2.31	.909	864	20.75	122	2.931	6.04	2.378	2.27	.894	115	7.018	23.6	1.440	150 x 75 x 6.5
24.0	7.32	16.13	2.31	.909	1060	25.47	151	3.628	5.87	2.311	2.22	.874	141	8.604	29.1	1.776	150 x 75 x 9
13.4	4.09	9.004	1.94	.764	425	10.21	65.5	1 571	4.99	1.965	1 96	.772	68.0	4.150	14.4	.879	125 x 65 x 6
9.36	2.85	6.290	1.55	.610	189	4.541	26.9	.646	3.98	1.567	1.50	.591	37.8	2.307	7.82	.477	100 x 50 x 5
6.92	2.11	4.650	1.27	.500	75.9	1.823	12.4	.298	2.93	1.154	1.19	.469	20.2	1.233	4.54	.277	75 x 40 x 5

Channels



IMPERIAL SERIES

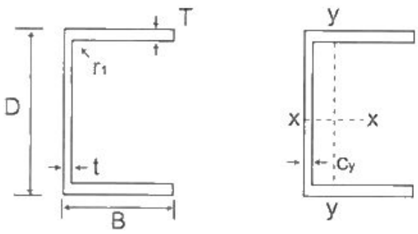
Designation		Depth of Section A	Width of Section B	Thickness		Distance of Cy	Root radius r_1	Toe radius r_2	Web	Ratios for Local Buckling		Moment of inertia	
Serial size	Mass per metre			Web t	Flange T				Depth between fillets d	Flange b/T	Web d/t	Axis x-x	Axis y-y
mm	kg	mm	mm	mm	mm	cm	mm	mm	mm			cm4	cm4
432 x 102	65.54	431.8	101.6	12.2	16.8	2.32	15.2	4.8	362.5	6.05	29.7	21399	628.6
381 x 102	55.10	381.0	101.6	10.4	16.3	2.52	15.2	4.8	312.6	6.23	30.0	14894	579.7
305 x 102	46.18	304.8	101.6	10.2	14.8	2.66	15.2	4.8	239.3	6.86	23.5	8214	499.5
305 x 89	41.69	304.8	88.9	10.2	13.7	2.18	13.7	3.2	245.4	6.49	24.1	7061	325.4
254 x 89	35.74	254.0	88.9	9.1	13.6	2.42	13.7	3.2	194.7	6.54	21.4	4448	302.4
254 x 76	28.29	254.0	75.2	8.1	10.9	1.86	12.2	3.2	203.9	6.99	25.2	3367	162.6
229 x 89	32.76	228.6	88.9	8.6	13.3	2.53	13.7	3.2	169.9	6.68	19.7	3387	285.0
229 x 76	26.06	228.6	76.2	11.2	2.00	12.2	3.2	177.8	6.80	23.4	2610	158.7	
203 x 89	29.76	203.2	88.9	8.1	12.9	2.65	13.7	3.2	145.2	6.89	17.9	2491	264.4
203 x 76	23.82	203.2	76.2	7.1	11.2	2.13	12.2	3.2	152.4	6.80	21.5	1950	151.3
178 x 89	26.81	177.8	88.9	7.6	12.3	2.76	13.7	3.2	121.0	7.23	15.9	1753	241.0
178 x 76	20.84	177.8	76.2	6.6	10.3	2.20	12.2	3.2	128.8	7.40	19.5	1337	134.0
152 x 89	23.84	152.4	88.9	7.1	11.6	2.86	13.7	3.2	96.9	7.66	13.7	1166	215.1
152 x 76	17.88	152.4	76.2	6.4	9.0	2.21	12.2	2.4	105.9	8.47	16.5	851.5	113.8
127 x 64	14.90	127.0	63.5	6.4	9.2	1.94	10.7	2.4	84.0	6.90	13.1	482.5	67.23
100 x 50	7.36	100.0	50.0	5.0	7.5	1.55	8	-	-	-	-	189	26.9
75 x 40	6.92	75.0	40.0	5.0	7.0	1.57	8	4	-	-	-	75.9	12.4



IMPERIAL SERIES

Radius of gyration		Elastic modulus		Plastic modulus							Designation	
Axis x-x	Axis y-y	Axis x-x	Axis y-y	Axis x-x	Axis y-y	Buckling Parameter u	Torsional Index x	Warping Constant H	Torsional Constant J	Area of Section	Mass per metre	Serial size
cm	cm	cm^3	cm^3	cm^3	cm^3			cm^6	cm	cm^2	kg	mm
16.0	2.74	991.1	80.14	1207	153.1	0.876	24.6	0.217	61.0	83.49	65.54	432 x 102
14.6	2.87	781.8	75.86	932.7	144.4	0.895	22.7	0.153	46.0	70.19	55.10	381 x 102
11.8	2.91	1539.0	66.59	638.3	128.1	0.900	18.9	0.0842	355.4	58.83	46.18	305 x 102
11.5	2.48	463.3	48.49	557.1	92.60	0.887	20.4	0.0551	27.6	53.11	41.69	305 x 89
9.88	2.58	350.2	46.70	414.4	89.56	0.906	17.1	0.0347	22.9	45.52	35.74	254 x 89
9.67	2.12	265.1	28.21	317.4	54.14	0.886	21.2	0.0194	12.3	36.03	28.29	254 x 76
9.01	2.61	296.4	44.82	348.4	86.38	0.912	15.5	0.0263	20.4	41.73	32.76	229 x 89
8.87	2.19	228.3	28.22	270.3	54.24	0.900	18.8	0.0151	11.4	33.20	26.06	229 x 76
8.10	2.64	245.2	42.34	266.6	81.62	0.915	14.1	0.0192	17.8	37.94	29.78	203 x 89
8.02	2.23	192.0	27.59	225.2	53.32	0.911	16.7	0.0112	10.4	30.34	23.82	203 x 76
7.16	2.66	197.2	39.29	229.6	75.44	0.915	12.7	0.0134	15.1	34.15	26.81	178 x 89
7.10	2.25	150.4	24.72	175.4	48.07	0.911	15.5	0.00764	8.13	26.54	20.84	178 x 76
6.20	2.66	153.0	35.70	177.7	68.12	0.909	11.3	0.00881	12.4	30.36	23.84	152 x 89
6.11	2.24	111.8	21.05	130.0	41.26	0.902	14.5	0.00486	5.94	22.77	17.8	152 x 76
5.04	1.88	75.99	15.25	89.4	29.31	0.910	11.7	0.00187	4.92	18.98	14.90	127 x 64
3.98	1.50	37.8	7.82	-	-	-	-	-	-	11.92	9.36	100 x 50
2.93	1.19	20.2	4.54	-	-	-	-	-	-	8.21	5.92	75 x 40

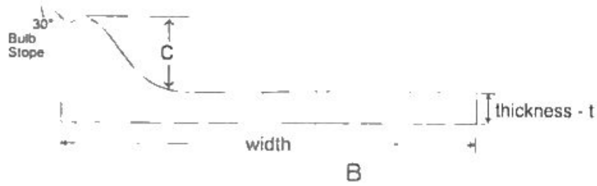
Parallel Flange Channels



Designation Size DxB	Mass Per Metre	Thickness Web t	Thickness Flange T	Root Radius r_1	Depth Between Fillets d	Area Of Section A	Centre Of Gravity Cy	Ratios For Local Buckling Flange	Ratios For Local Buckling Web	Second Moment Of Area Axis x-x	Second Moment Of Area Axis y-y
	kg/m	mm	mm	mm	mm	cm²	cm	B/T	d/t	cm⁴	cm⁴
100x50	10.2	5.0	8.5	9	65.0	13.0	1.73	5.88	13.0	208	32.3
125x65	14.8	5.5	9.5	12	82.0	18.8	2.25	6.84	14.9	483	80.0
150x75	17.9	5.5	10.0	12	106	22.8	2.58	7.50	19.3	861	131
150x90	23.9	6.5	12.0	12	102	30.4	3.30	7.50	15.7	1162	253
180x75	20.3	6.0	10.5	12	135	25.9	2.41	7.14	22.5	1370	146
180x90	26.1	6.5	12.5	12	131	33.2	3.17	7.20	20.2	1817	277
200x75	23.4	6.0	12.5	12	151	29.9	2.48	6.00	25.2	1963	170
200x90	29.7	7.0	14.0	12	148	37.9	3.12	6.43	21.1	2523	314
230x75	25.7	6.5	12.5	12	181	32.7	2.30	6.00	27.8	2748	181
230x90	32.2	7.5	14.0	12	178	41.0	2.92	6.43	23.7	3518	334
250x90	35.5	8.0	15.0	12	220	45.2	2.86	6.00	27.5	4510	364
260x75	27.6	7.0	12.0	12	212	35.1	2.10	6.25	30.3	3619	185
260x90	34.8	8.0	14.0	12	208	44.4	2.74	6.43	26.0	4728	353
300x90	41.4	9.0	15.5	12	245	52.7	2.60	5.81	27.2	7218	404
300x100	45.5	9.0	16.5	15	237	58.0	3.05	6.06	26.3	8229	568
380x100	54.0	9.5	17.5	15	315	68.7	2.79	5.71	33.2	15030	643
430x100	64.4	11.0	19.0	15	362	82.1	2.62	5.26	32.9	21940	722

Designation Size DxB	Mass Per Metre	Radius Of Gyration Axis x-x	Radius Of Gyration Axis y-y	Elastic Modulus Axis x-x	Elastic Modulus Axis y-y	Plastic Modulus Axis x-x	Plastic Modulus Axis y-y	Buckling Parameter u	Torsional Index x	Warping Constant H	Torsional Constant J
	kg/m	cm	cm	cm³	cm³	cm³	cm³			dm⁶	cm⁴
100x50	10.2	4.00	1.58	41.5	9.89	48.9	17.5	0.942	10.0	0.000491	2.53
125x65	14.8	5.07	2.06	77.3	18.8	89.9	33.2	0.942	11.1	0.00194	4.72
150x75	17.9	6.15	2.40	115	26.6	132	47.2	0.946	13.1	0.00467	6.10
150x90	23.9	6.18	2.89	155	44.4	179	76.9	0.936	10.8	0.00890	11.8
180x75	20.3	7.27	2.38	152	28.8	176	51.8	0.946	15.3	0.00754	7.34
180x90	26.1	7.40	2.89	202	47.4	232	83.5	0.949	12.8	0.0141	13.3
200x75	23.4	8.11	2.39	196	33.8	227	60.6	0.956	14.8	0.0107	11.1
200x90	29.7	8.16	2.89	252	53.4	291	94.5	0.954	12.9	0.0197	13.3
230x75	25.7	9.17	2.35	239	34.8	278	65.2	0.947	17.3	0.0153	11.8
230x90	32.2	9.27	2.86	306	55.0	355	98.9	0.950	15.1	0.0279	19.3
250x90	35.5	9.99	2.84	361	59.3	421	107	0.948	15.5	0.0359	23.8
260x75	27.6	10.1	2.30	278	34.4	328	62.0	0.932	20.5	0.0203	11.7
260x90	34.8	10.3	2.82	364	56.3	425	102	0.942	17.2	0.0379	20.6
300x90	41.4	11.7	2.77	481	63.1	568	114	0.934	18.4	0.0581	28.8
300x100	45.5	11.9	3.13	549	81.7	641	148	0.944	17.0	0.0813	36.8
380x100	54.0	14.8	3.06	791	89.2	933	161	0.932	21.2	0.150	45.7
430x100	64.4	16.3	2.97	1020	97.9	1222	176	0.917	22.5	0.219	63.0

Bulb Flats



Width B mm	Thickness t mm	Bulb Height C mm	Area of Section A mm^2	Mass Per Metre kg/m
120	6.0	17.0	9.31	7.31
	7.0	17.0	10.5	8.25
	8.0	17.0	11.7	9.19
140	6.5	19.0	11.7	9.21
	7.0	19.0	12.4	9.74
	8.0	19.0	13.8	10.83
	10.0	19.0	16.6	13.03
160	7.0	22.0	14.6	11.4
	8.0	22.0	16.2	12.7
	9.0	22.0	17.8	14.0
	11.5	22.0	21.8	17.3
180	8.0	25.0	18.9	14.8
	9.0	25.0	20.7	16.2
	10.0	25 0	22.5	17.6
	11.5	25.0	25.2	19.7
200	8.5	28.0	22 6	17.8
	9.0	28.0	23.6	18.5
	10.0	28.0	25.6	20.1
	11.0	28.0	27.6	21.7
	12.0	28.0	29.6	23.2
220	9.0	31.0	26.8	21.0
	10.0	31.0	29.0	22.8
	11.0	31.0	31.2	24.5
	12.0	31.0	33.4	26.2
240	9 5	34 0	31.2	24.4
	10.0	34.0	32.4	25.4
	11.0	34.0	34.9	27.4
	12.0	34.0	37 3	29.3
260	10.0	37.0	36.1	28.3
	11.0	37.0	38.7	30.3
	12.0	37.0	41.3	32.4
280	10.5	40.0	41.2	32.4
	11.0	40.0	42.6	33.5
	12.0	40.0	45.5	35.7
	13.0	40.0	48.4	37.9
300	11.0	43.0	46 7	36.7
	12.0	43.0	49.7	39.0
	13.0	43.0	52.8	41.5
320	11.5	46.0	52.6	41.2
	12.0	46.0	54.2	42.5
	13.0	46.0	57.4	45.0
	14.0	46.0	60.6	47.5
340	12.0	49.0	58.8	46.1
	13.0	49.0	62.2	48.8
	14.0	49.0	65.5	51 5
	15.0	49.0	69.0	54.2
370	12.5	53.5	67.8	53.1
	13.0	53.5	69.6	54.6
	14.0	53.5	73.3	57.5
	15.0	53.5	77.0	60.5
	16.0	53.5	80.7	63.5
400	13.0	58.0	77.4	60.8
	14.0	58 0	81 4	63.9
	15.0	58.0	85.4	67.0
	16.0	58.0	89.4	70.2
430	14.0	62.5	89.7	70.6
	15.0	62.5	94.1	73.9
	17.0	62.5	103.0	80.6
	20.0	62.5	115.0	90.8